

Sorting Visualization Steps

Step 1 of 78

Starting sorting visualization...

Array: [64, 34, 25, 12, 22, 11, 90, 88, 76, 50]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
64	34	25	12	22	11	90	88	76	50

Step 2 of 78

Comparing elements at positions 0 and 1

Array: [64, 34, 25, 12, 22, 11, 90, 88, 76, 50]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
64	34	25	12	22	11	90	88	76	50

Step 3 of 78

Swapping elements at positions 0 and 1

Array: [34, 64, 25, 12, 22, 11, 90, 88, 76, 50]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
34	64	25	12	22	11	90	88	76	50

Step 4 of 78

Comparing elements at positions 1 and 2

Array: [34, 64, 25, 12, 22, 11, 90, 88, 76, 50]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
34	64	25	12	22	11	90	88	76	50

Step 5 of 78

Swapping elements at positions 1 and 2

Array: [34, 25, 64, 12, 22, 11, 90, 88, 76, 50]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
34	25	64	12	22	11	90	88	76	50

Step 6 of 78

Comparing elements at positions 2 and 3

Array: [34, 25, 64, 12, 22, 11, 90, 88, 76, 50]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
34	25	64	12	22	11	90	88	76	50

Step 7 of 78

Swapping elements at positions 2 and 3

Array: [34, 25, 12, 64, 22, 11, 90, 88, 76, 50]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
34	25	12	64	22	11	90	88	76	50

Step 8 of 78

Comparing elements at positions 3 and 4

Array: [34, 25, 12, 64, 22, 11, 90, 88, 76, 50]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
34	25	12	64	22	11	90	88	76	50

Step 9 of 78

Swapping elements at positions 3 and 4

Array: [34, 25, 12, 22, 64, 11, 90, 88, 76, 50]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
34	25	12	22	64	11	90	88	76	50

Step 10 of 78

Comparing elements at positions 4 and 5

Array: [34, 25, 12, 22, 64, 11, 90, 88, 76, 50]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
34	25	12	22	64	11	90	88	76	50

Step 11 of 78

Swapping elements at positions 4 and 5

Array: [34, 25, 12, 22, 11, 64, 90, 88, 76, 50]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
34	25	12	22	11	64	90	88	76	50

Step 12 of 78

Comparing elements at positions 5 and 6

Array: [34, 25, 12, 22, 11, 64, 90, 88, 76, 50]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
34	25	12	22	11	64	90	88	76	50

Step 13 of 78

Comparing elements at positions 6 and 7

Array: [34, 25, 12, 22, 11, 64, 90, 88, 76, 50]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
34	25	12	22	11	64	90	88	76	50

Step 14 of 78

Swapping elements at positions 6 and 7

Array: [34, 25, 12, 22, 11, 64, 88, 90, 76, 50]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
34	25	12	22	11	64	88	90	76	50

Step 15 of 78

Comparing elements at positions 7 and 8

Array: [34, 25, 12, 22, 11, 64, 88, 90, 76, 50]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
34	25	12	22	11	64	88	90	76	50

Step 16 of 78

Swapping elements at positions 7 and 8

Array: [34, 25, 12, 22, 11, 64, 88, 76, 90, 50]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
34	25	12	22	11	64	88	76	90	50

Step 17 of 78

Comparing elements at positions 8 and 9

Array: [34, 25, 12, 22, 11, 64, 88, 76, 90, 50]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
34	25	12	22	11	64	88	76	90	50

Step 18 of 78

Swapping elements at positions 8 and 9

Array: [34, 25, 12, 22, 11, 64, 88, 76, 50, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
34	25	12	22	11	64	88	76	50	90

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1 elements sorted

Array: [34, 25, 12, 22, 11, 64, 88, 76, 50, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
34	25	12	22	11	64	88	76	50	90

Step 20 of 78

Comparing elements at positions 0 and 1

Array: [34, 25, 12, 22, 11, 64, 88, 76, 50, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
34	25	12	22	11	64	88	76	50	90

Step 21 of 78

Swapping elements at positions 0 and 1

Array: [25, 34, 12, 22, 11, 64, 88, 76, 50, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
25	34	12	22	11	64	88	76	50	90

Step 22 of 78

Comparing elements at positions 1 and 2

Array: [25, 34, 12, 22, 11, 64, 88, 76, 50, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
25	34	12	22	11	64	88	76	50	90

Step 23 of 78

Swapping elements at positions 1 and 2

Array: [25, 12, 34, 22, 11, 64, 88, 76, 50, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
25	12	34	22	11	64	88	76	50	90

Step 24 of 78

Comparing elements at positions 2 and 3

Array: [25, 12, 34, 22, 11, 64, 88, 76, 50, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
25	12	34	22	11	64	88	76	50	90

Step 25 of 78

Swapping elements at positions 2 and 3

Array: [25, 12, 22, 34, 11, 64, 88, 76, 50, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
25	12	22	34	11	64	88	76	50	90

Step 26 of 78

Comparing elements at positions 3 and 4

Array: [25, 12, 22, 34, 11, 64, 88, 76, 50, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
25	12	22	34	11	64	88	76	50	90

Step 27 of 78

Swapping elements at positions 3 and 4

Array: [25, 12, 22, 11, 34, 64, 88, 76, 50, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
25	12	22	11	34	64	88	76	50	90

Step 28 of 78

Comparing elements at positions 4 and 5

Array: [25, 12, 22, 11, 34, 64, 88, 76, 50, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
25	12	22	11	34	64	88	76	50	90

Step 29 of 78

Comparing elements at positions 5 and 6

Array: [25, 12, 22, 11, 34, 64, 88, 76, 50, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
25	12	22	11	34	64	88	76	50	90

Step 30 of 78

Comparing elements at positions 6 and 7

Array: [25, 12, 22, 11, 34, 64, 88, 76, 50, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
25	12	22	11	34	64	88	76	50	90

Step 31 of 78

Swapping elements at positions 6 and 7

Array: [25, 12, 22, 11, 34, 64, 76, 88, 50, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
25	12	22	11	34	64	76	88	50	90

Step 32 of 78

Comparing elements at positions 7 and 8

Array: [25, 12, 22, 11, 34, 64, 76, 88, 50, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
25	12	22	11	34	64	76	88	50	90

Step 33 of 78

Swapping elements at positions 7 and 8

Array: [25, 12, 22, 11, 34, 64, 76, 50, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
25	12	22	11	34	64	76	50	88	90

Step 34 of 78

2 elements sorted

Array: [25, 12, 22, 11, 34, 64, 76, 50, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
25	12	22	11	34	64	76	50	88	90

Step 35 of 78

Comparing elements at positions 0 and 1

Array: [25, 12, 22, 11, 34, 64, 76, 50, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
25	12	22	11	34	64	76	50	88	90

Step 36 of 78

Swapping elements at positions 0 and 1

Array: [12, 25, 22, 11, 34, 64, 76, 50, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
12	25	22	11	34	64	76	50	88	90

Step 37 of 78

Comparing elements at positions 1 and 2

Array: [12, 25, 22, 11, 34, 64, 76, 50, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
12	25	22	11	34	64	76	50	88	90

Step 38 of 78

Swapping elements at positions 1 and 2

Array: [12, 22, 25, 11, 34, 64, 76, 50, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
12	22	25	11	34	64	76	50	88	90

Step 39 of 78

Comparing elements at positions 2 and 3

Array: [12, 22, 25, 11, 34, 64, 76, 50, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
12	22	25	11	34	64	76	50	88	90

Step 40 of 78

Swapping elements at positions 2 and 3

Array: [12, 22, 11, 25, 34, 64, 76, 50, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
12	22	11	25	34	64	76	50	88	90

Step 41 of 78

Comparing elements at positions 3 and 4

Array: [12, 22, 11, 25, 34, 64, 76, 50, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
12	22	11	25	34	64	76	50	88	90

Step 42 of 78

Comparing elements at positions 4 and 5

Array: [12, 22, 11, 25, 34, 64, 76, 50, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
12	22	11	25	34	64	76	50	88	90

Step 43 of 78

Comparing elements at positions 5 and 6

Array: [12, 22, 11, 25, 34, 64, 76, 50, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
12	22	11	25	34	64	76	50	88	90

Step 44 of 78

Comparing elements at positions 6 and 7

Array: [12, 22, 11, 25, 34, 64, 76, 50, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
12	22	11	25	34	64	76	50	88	90

Step 45 of 78

Swapping elements at positions 6 and 7

Array: [12, 22, 11, 25, 34, 64, 50, 76, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
12	22	11	25	34	64	50	76	88	90

Step 46 of 78

3 elements sorted

Array: [12, 22, 11, 25, 34, 64, 50, 76, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
12	22	11	25	34	64	50	76	88	90

Step 47 of 78

Comparing elements at positions 0 and 1

Array: [12, 22, 11, 25, 34, 64, 50, 76, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
12	22	11	25	34	64	50	76	88	90

Step 48 of 78

Comparing elements at positions 1 and 2

Array: [12, 22, 11, 25, 34, 64, 50, 76, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
12	22	11	25	34	64	50	76	88	90

Step 49 of 78

Swapping elements at positions 1 and 2

Array: [12, 11, 22, 25, 34, 64, 50, 76, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
12	11	22	25	34	64	50	76	88	90

Step 50 of 78

Comparing elements at positions 2 and 3

Array: [12, 11, 22, 25, 34, 64, 50, 76, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
12	11	22	25	34	64	50	76	88	90

Step 51 of 78

Comparing elements at positions 3 and 4

Array: [12, 11, 22, 25, 34, 64, 50, 76, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
12	11	22	25	34	64	50	76	88	90

Step 52 of 78

Comparing elements at positions 4 and 5

Array: [12, 11, 22, 25, 34, 64, 50, 76, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
12	11	22	25	34	64	50	76	88	90

Step 53 of 78

Comparing elements at positions 5 and 6

Array: [12, 11, 22, 25, 34, 64, 50, 76, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
12	11	22	25	34	64	50	76	88	90

Step 54 of 78

Swapping elements at positions 5 and 6

Array: [12, 11, 22, 25, 34, 50, 64, 76, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
12	11	22	25	34	50	64	76	88	90

Step 55 of 78

4 elements sorted

Array: [12, 11, 22, 25, 34, 50, 64, 76, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
12	11	22	25	34	50	64	76	88	90

Step 56 of 78

Comparing elements at positions 0 and 1

Array: [12, 11, 22, 25, 34, 50, 64, 76, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
12	11	22	25	34	50	64	76	88	90

Step 57 of 78

Swapping elements at positions 0 and 1

Array: [11, 12, 22, 25, 34, 50, 64, 76, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
11	12	22	25	34	50	64	76	88	90

Step 58 of 78

Comparing elements at positions 1 and 2

Array: [11, 12, 22, 25, 34, 50, 64, 76, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
11	12	22	25	34	50	64	76	88	90

Step 59 of 78

Comparing elements at positions 2 and 3

Array: [11, 12, 22, 25, 34, 50, 64, 76, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
11	12	22	25	34	50	64	76	88	90

Step 60 of 78

Comparing elements at positions 3 and 4

Array: [11, 12, 22, 25, 34, 50, 64, 76, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
11	12	22	25	34	50	64	76	88	90

Step 61 of 78

Comparing elements at positions 4 and 5

Array: [11, 12, 22, 25, 34, 50, 64, 76, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
11	12	22	25	34	50	64	76	88	90

Step 62 of 78

5 elements sorted

Array: [11, 12, 22, 25, 34, 50, 64, 76, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
11	12	22	25	34	50	64	76	88	90

Step 63 of 78

Comparing elements at positions 0 and 1

Array: [11, 12, 22, 25, 34, 50, 64, 76, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
11	12	22	25	34	50	64	76	88	90

Step 64 of 78

Comparing elements at positions 1 and 2

Array: [11, 12, 22, 25, 34, 50, 64, 76, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
11	12	22	25	34	50	64	76	88	90

Step 65 of 78

Comparing elements at positions 2 and 3

Array: [11, 12, 22, 25, 34, 50, 64, 76, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
11	12	22	25	34	50	64	76	88	90

Step 66 of 78

Comparing elements at positions 3 and 4

Array: [11, 12, 22, 25, 34, 50, 64, 76, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
11	12	22	25	34	50	64	76	88	90

Step 67 of 78

6 elements sorted

Array: [11, 12, 22, 25, 34, 50, 64, 76, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
11	12	22	25	34	50	64	76	88	90

Step 68 of 78

Comparing elements at positions 0 and 1

Array: [11, 12, 22, 25, 34, 50, 64, 76, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
11	12	22	25	34	50	64	76	88	90

Step 69 of 78

Comparing elements at positions 1 and 2

Array: [11, 12, 22, 25, 34, 50, 64, 76, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
11	12	22	25	34	50	64	76	88	90

Step 70 of 78

Comparing elements at positions 2 and 3

Array: [11, 12, 22, 25, 34, 50, 64, 76, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
11	12	22	25	34	50	64	76	88	90

Step 71 of 78

7 elements sorted

Array: [11, 12, 22, 25, 34, 50, 64, 76, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
11	12	22	25	34	50	64	76	88	90

Step 72 of 78

Comparing elements at positions 0 and 1

Array: [11, 12, 22, 25, 34, 50, 64, 76, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
11	12	22	25	34	50	64	76	88	90

Step 73 of 78

Comparing elements at positions 1 and 2

Array: [11, 12, 22, 25, 34, 50, 64, 76, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
11	12	22	25	34	50	64	76	88	90

Step 74 of 78

8 elements sorted

Array: [11, 12, 22, 25, 34, 50, 64, 76, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
11	12	22	25	34	50	64	76	88	90

Step 75 of 78

Comparing elements at positions 0 and 1

Array: [11, 12, 22, 25, 34, 50, 64, 76, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
11	12	22	25	34	50	64	76	88	90

Step 76 of 78

9 elements sorted

Array: [11, 12, 22, 25, 34, 50, 64, 76, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
11	12	22	25	34	50	64	76	88	90

Step 77 of 78

10 elements sorted

Array: [11, 12, 22, 25, 34, 50, 64, 76, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
11	12	22	25	34	50	64	76	88	90

Step 78 of 78

Array is fully sorted!

Array: [11, 12, 22, 25, 34, 50, 64, 76, 88, 90]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
11	12	22	25	34	50	64	76	88	90